

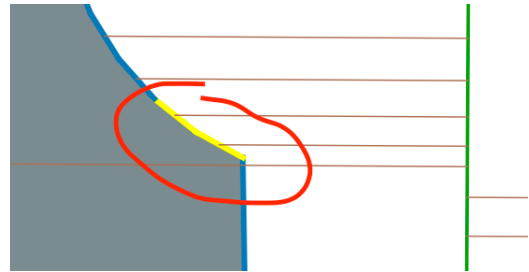
Question 1

Objective

To determine if it worthwhile to implement an algorithm to improve street representation for the lot line labelling algorithm.

Attempts

The current implementation of the lot line labelling algorithm, found in the link below, misidentifies some oblique fronts as sides due to misrepresenting the street as a line instead of a polygon.



Evidence

Report on how to convert streets from lines to polygons: https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/blob/main/code/gis_files/reports/improving_street_representation.pdf

Current lot line labelling implementation: https://github.com/ubco-mds-2025-labs/w2025-data599-capstone-projects-green-metrics-technology/tree/main/code/gis_files/burnaby

Diagnosis

Since municipal files store streets as lines instead of polygons, the lot line labelling algorithm thinks some front lot lines are more oblique than they really are, causing them to be mislabelled.

Proposed Next Steps

Based on the current result for 1000 Burnaby parcels, mislabelled fronts are a small issue, occurring in less than 1% of parcels. Due to time restrictions for the capstone project, we suggest moving forward to define buildable area with the current version of the lot line labelling algorithm.

Decision Ask

Do you think implementing the algorithm to convert streets from lines to polygons is worthwhile considering it could be difficult to implement with marginal benefit?